

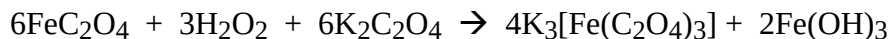
***Preparation & Analysis of potassium
trioxalato ferrate (III) trihydrate***
 $K_3[Fe(C_2O_4)_3] \cdot 3H_2O$

Step One:**Preparation of Ferrous Oxalate**

Weigh the vial containing the sample of ferrous ammonium sulfate, $(NH_4)_2SO_4 \cdot FeSO_4 \cdot 6H_2O$. Transfer the ferrous ammonium sulfate to a beaker containing 25 mL of warm water acidified with 1 mL dilute 3M H_2SO_4 . Dry the vial and weigh the empty vial. **Record the mass of ferrous ammonium sulfate used.** To this solution add 25 mL of oxalic acid solution (5 g/50 mL). **Cautiously** heat the solution to boiling on a hot plate, continuously stirring the solution to avoid bumping. Remove and then allow the precipitate to settle. Decant and discard the clear supernatant liquid (avoid loss of precipitate). Stir the remaining precipitate with 25 mL of hot water, the solubility of $FeC_2O_4 \cdot 2H_2O$ is .022 g/100 mL in cold water and .026 g/100 mL in hot water. Decant and discard the wash liquid. Add 25 mL of hot water to the precipitate and stir, then filter by suction using a small Büchner funnel and wash the precipitate thoroughly with small portions (10-20 mL) of hot water. Finally rinse with 20 mL of acetone. **Weigh the final product.**

Step Two:**Preparation of $K_3[Fe(C_2O_4)_3] \cdot 3H_2O$**

Suspend the washed and dried ferrous oxalate in 30 mL of warm solution containing 3.5 g of potassium oxalate ($K_2C_2O_4 \cdot H_2O$). Place this solution in an ice bath and add **slowly** with continuous stirring 7-10 mL of 30% H_2O_2 **in very small portions**, then heat the mixture to boiling and dissolve the precipitate by adding in one portion 7 mL of a solution of oxalic acid (containing 2.5 g/25 mL). A further 3 mL is then added drop by drop using an eye dropper (an excess of oxalic acid is to be avoided). The liquid should be near the boiling point while these additions are being made.



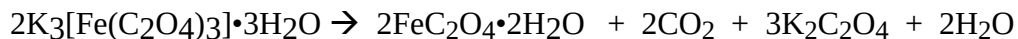
30% H_2O_2 is a strong oxidizer, a corrosive irritant to skin, eyes, mucous membranes and should be handled with care.

18M H_2SO_4 sulfuric acid is a strong acid, corrosive irritant and contact will result in rapid destruction of tissues, causing severe burns.

Filter through a Büchner funnel. Transfer the hot solution to a clean beaker and while still hot, add to the filtrate 15 mL of 95% ethanol and redissolve any precipitated crystals by gently heating. Then store the solution **in a dark cupboard** for crystallization.

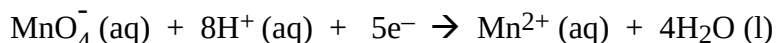
After crystallization has occurred, filter with suction and wash the product on the filter paper with 20 mL of an equi-volume mixture of ethanol and water and finally with 20 mL of acetone. Draw air through the precipitate for several minutes. Weigh the dry crystals.

$\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$ is highly photosensitive. The bright green crystals on exposure soon become covered with a yellow powder of ferrous oxalate.



Oxidation - Reduction Titration

Potassium permanganate (KMnO_4) is commonly used as an oxidizing agent and can be reduced in different ways, depending on the H^+ concentration of the reaction mixture. In neutral or basic solutions a mixture of products may result, chiefly solid MnO_2 and green manganate ion, MnO_4^{2-} . In acid solution, however, the reduction goes by one course only, the half-reaction being:



Part A Preparation of a 0.02 M KMnO_4 Solution

Handle solid KMnO_4 and the solution carefully, for if either gets on clothes or skin, it will be reduced to MnO_2 leaving brown spots.

Traces of MnO_2 initially present in the permanganate are formed by the reaction of permanganate with residual organic impurities (reducing material) in the water will catalyze the reduction of MnO_4^- .

Weigh out approximately 1.2 - 1.40 g of potassium permanganate and add it to approximately 350 mL of boiled distilled water in a 600 mL beaker. Stir the solution well to dissolve all the crystals. Cover the solution with a watch glass and keep the solution near the boiling point for 20 minutes. Allow the solution to cool slightly, then filter the solution through a non-reducing filter (glass wool) placed in the neck of a short stemmed funnel. The KMnO_4 solution can be stored in a plastic bottle. Be sure the bottle is labelled with your name, lab number, semester and year.

Part B Standardization of KMnO_4 Solution

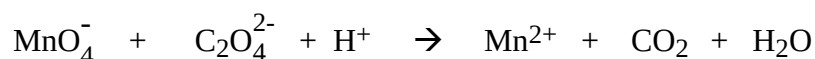
Note 1. The $\text{MnO}_4^- / \text{C}_2\text{O}_4^{2-}$ reaction is slow at room temperature and hence the titration is carried out at elevated temperatures. Even at higher temperatures the reaction starts slowly, but the rate increases as Manganous (Mn^{2+}) ion, which acts as a catalyst, is formed. Since the Mn^{2+} ion is a product of the reaction itself, the reaction is called "autocatalytic".

Note 2. With deeply colored solutions like permanganate, it may be necessary to read the top of the meniscus since it is difficult to see the bottom of it.

Weigh accurately, to the nearest tenth of a milligram, three samples each in the range .18 - .22 g of sodium oxalate ($\text{Na}_2\text{C}_2\text{O}_4$) into clean 250 mL Erlenmeyer flasks. To each sample add 60 mL of water and 10 mL of 3 M (dilute) sulfuric acid. **Titrate each solution as follows.**

Heat the contents of the Erlenmeyer flask to 80 - 90°C, but not over 90°C. (Do not warm all trials at the same time). Rinse the thermometer with 1 to 2 mL of distilled water and remove and titrate the hot solution with the KMnO_4 solution. Add several mL of KMnO_4 solution slowly to initiate the reaction then larger portions may be added until the endpoint is approached. Then slow the addition of titrant to avoid overshooting the faint pink endpoint. The solution should be swirled while additions are being made.

The reaction is represented by the following reactants and products **in acid solution**.



Balance the equation and calculate the molarity of your permanganate solution.

Characterization of the Green Crystals:

A. % H₂O in the green crystals

Weigh 0.7-1.0 g (± 0.0001 g) of the complex salt into a pre-weighed evaporating dish. Label or mark the dish so you can identify it and place the evaporating dish and complex salt in an oven set at 110°C for approximately 45 minutes. Cool the dish and contents and weigh to the nearest ± 0.1 mg. Calculate the % H₂O in the crystals. **Use the dried crystals for part C.**

B. % of Oxalate in the green crystals using Potassium Permanganate



Proceed with the **prepared salt** complex. Weigh out 3 samples of the complex salt in the weight range of .18 - .22 g (on the analytical balance) into clean 250 mL Erlenmeyer flasks. Dissolve each sample in 60 mL H₂O, 10 mL of 3 M (dilute) sulfuric acid and heat each sample to 80-90°C but not over 90° and titrate with standardized potassium permanganate until the first appearance of a faint permanent pink color occurs. Calculate the percent oxalate in the complex salt.

C. % of Iron in the green crystals

This procedure should be **carried out in the fumehood** until the final step. To the evaporating dish and dried crystals, add cautiously approximately 3 mL concentrated H₂SO₄ (sulfuric acid) and cover with a watch glass. **In the fumehood**, using a hotplate, gently heat the dish and contents for several minutes until the bubbling resulting from the decomposition of the oxalate ion ceases rather abruptly. Allow the evaporating dish and contents to cool in the hood.

Then pour the colorless (anhydrous) ferric sulfate from the dish into an Erlenmeyer flask and rinse the watch glass. Add approximately 25 mL of distilled H₂O to the Erlenmeyer flask and heat the flask and contents to boiling. The milky solution will gradually turn to a clear pale yellow solution. Then add small portions of zinc dust and continue to cautiously heat the flask until the yellow color disappears from the solution. Heat for approximately 1 minute beyond this point. The hot mixture is filtered through #1 Whatman paper into a clean flask and rinsed to achieve complete transfer. This solution is then titrated with standardized KMnO₄ solution until a faint pink/orange endpoint appears.

The % Fe in the green crystals can then be **calculated**. Use the original mass of the green crystals **before they were dried in the oven**.

Preparation and Analysis of Potassium trioxalato ferrate (III) trihydrate

Name: _____

Lab Section# : _____

Data Sheet:

1. Step one: Preparation of Ferrous oxalate

Ferrous Ammonium Sulfate sample # _____

Mass of Ferrous ammonium sulfate•hexahydrate used _____ g

Mass of 250mL beaker _____ g

Instructors' signature: _____

2. Step two: Preparation of Potassium trioxalato ferrate (III) trihydrate

Mass of ferrous oxalate •dihydrate used to start the reaction _____ g

Mass of Potassium Tris-oxalatoferrate III trihydrate produced _____ g
(Green Crystals)**Part B: Standardization of the prepared KMnO_4 solution**

	Trial #1	Trial #2	Trial #3	Trial #4
Mass $\text{Na}_2\text{C}_2\text{O}_4$ & container (g)	_____	_____	_____	_____
Mass container (g)	_____	_____	_____	_____
Mass of $\text{Na}_2\text{C}_2\text{O}_4$ used(g)	_____	_____	_____	_____
Initial Volume KMnO_4	_____	_____	_____	_____
Final Volume KMnO_4	_____	_____	_____	_____
Volume of KMnO_4 used (mL)	_____	_____	_____	_____
Molarity of KMnO_4	_____ M,	_____ M,	_____ M,	_____ M
Average Molarity of KMnO_4	_____ M (**Report this value on p.42**)			

Instructors' signature: _____

Preparation and Analysis of Potassium trioxalato ferrate (III) trihydrate

Name: _____ Date: _____

Analysis of the of Potassium trioxalato ferrate (III) trihydrate**A. % H₂O in the Green Crystals:**

Mass of the evaporating dish _____g

Mass of green crystals & the evaporating dish **before** heating _____gMass of green crystals **before** heating _____gMass of green crystals & the evaporating dish **after** heating _____gMass of green crystals **after** heating _____g**B. % oxalate ion in the Green Crystals:**

	Trial #1	Trial #2	Trial #3	Trial #4
Mass crystals & container (g)	_____	_____	_____	_____
Mass container (g)	_____	_____	_____	_____
Mass green crystals used(g)	_____	_____	_____	_____
Initial Volume KMnO₄	_____	_____	_____	_____
Final Volume KMnO₄	_____	_____	_____	_____
Volume of KMnO₄ used (mL)	_____	_____	_____	_____

C. % of Iron in the Green Crystals:Mass of green crystals **before** heating _____gInitial Volume **KMnO₄** _____ mLFinal Volume **KMnO₄** _____ mLVolume of **KMnO₄** used (mL) _____ mL

Instructors' signature: _____

Experiment #2 **Preparation of $K_3[Fe(C_2O_4)_3] \cdot 3H_2O$**

Report Sheet:

Name: _____

Sample # _____

Date: _____

Show example calculations on the back of the report sheets or append separate sheets.

1. Balanced Equations:

Step #1.

Step #2. (The equation is in the manual, write a balanced equation in the hydrated form)

2. Theoretical Yield (Show your work): Using your experimental mass of starting material for each calculation.

Step #1.

Step #2.

3. Per Cent Yield (Show your work):

Step #1.

Step #2.

Experiment #2 Characterization of $K_3[Fe(C_2O_4)_3] \cdot 3H_2O$

Report Sheet:

Name: _____

Lab Section: _____

Balanced titration and reaction equations for: Fe^{3+} & Zn; MnO_4^- & $C_2O_4^{2-}$; and Fe^{2+} & MnO_4^- .

1.

2.

3.

Molarity of $KMnO_4$: (From page 39).

Average Molarity of $KMnO_4$ = _____ M.

Experimental weight % of $C_2O_4^{2-}$ in the prepared green crystals:

Trial #1.

Trial #2.

Trial #3.

Trial #4.

Average weight % of $C_2O_4^{2-}$:

Experimental weight % of Fe in the prepared green crystals:

Trial #1:

Experimental weight % H₂O in the prepared green crystals:

Experimental weight % K in the prepared green crystals:

Formula Weight % of Fe, H₂O, C₂O₄²⁻, K in K₃[Fe(C₂O₄)₃] • 3H₂O:

% Deviation of the Experimental weight % values compared to Formula Weight % values:

Conclusions:

Preparation and Analysis of Potassium trioxalato ferrate (III) trihydrate**Example Pre-Lab Questions:**

1. Write the chemical formula of potassium permanganate.
2. A mixture of 5.00 g of $\text{H}_2(\text{g})$ and 10.00 g $\text{O}_2(\text{g})$ is ignited. Water forms according to the following reaction: $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{g})$ Which is the limiting reagent? How much water will be produced?
3. Name this compound: $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$
4. Write the chemical formula for ferrous oxalate dihydrate.
5. Describe the proper procedure for diluting 18 M concentrated sulfuric acid to prepare 100 mL of 3 M sulfuric acid.
6. A solution of potassium permanganate was prepared by dissolving 21.08 g of KMnO_4 in water and diluting to a total volume of 800mL. What is the molarity of the prepared solution?
7. Is oxalic acid a liquid or a solid under normal laboratory conditions?
8. Is oxidation or reduction occurring in the following reaction? Indicate why.

$$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$$
9. DDT (dichlorodiphenyltrichloroethane) is an insectide banned in Canada because it is hazardous to fish, birds and humans. The reaction for its preparation is;
 $2\text{C}_6\text{H}_5\text{Cl}$ (chlorobenzene) + C_2HOCl_3 (chloral) $\rightarrow \text{C}_{14}\text{H}_9\text{Cl}_5$ (DDT) + H_2O
 If 450 g of chlorobenzene reacts with 899 g of chloral calculate (a) the theoretical yield of DDT in grams and (b) the % yield if 650 g of DDT is actually isolated.
10. What color change occurs at the endpoint of the permanganate - oxalate reaction?
11. Why should potassium permanganate be handled carefully?
12. What is the catalyst used in the permanganate - oxalate reaction?
13. What is the color of ferrous oxalate?
14. What color are the crystals of $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$?
15. Why are you analyzing the crystals of $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$?
16. Explain the objective(s) in each of the 3 phases of a synthesis.